

# FRA Diagnostic Analysis Report

**Standard:** IEEE C57.149-2012 / IEC 60076-18 Standards-Aligned SFRA Diagnostic Pipeline | **System:** AI-Powered Hybrid Transformer Integrity Monitor

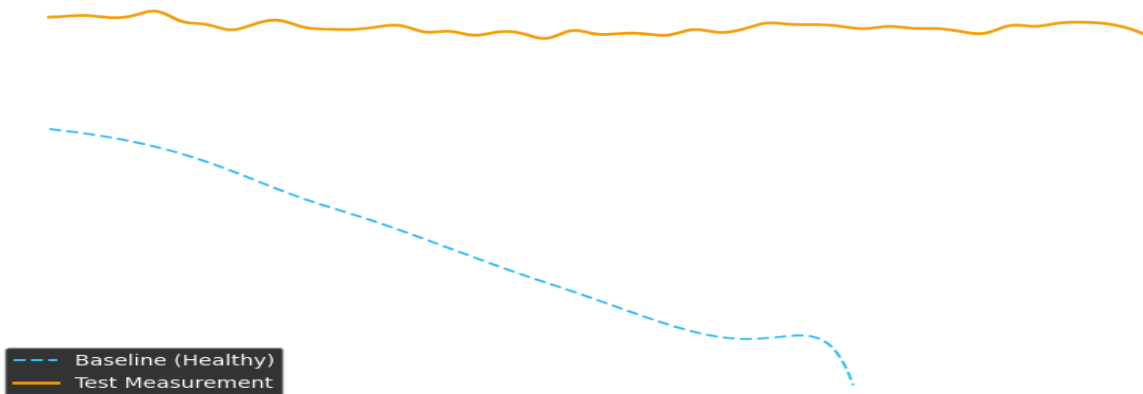
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|-------------------------------|--------------------------------------------|
| Asset / Transformer ID: TR005 | Inspection Date: 2026-03-05                |
| Winding Configuration: LV     | Overall Status / Severity: Critical (High) |

|                                                                    |       |                                                                                                                                                                  |
|--------------------------------------------------------------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Overall Health Integrity Score<br>Blended Physics (60%) + ML (40%) | 20.0% | Condition Assessment: Critical Fault Condition<br>AI Fault Signature: Winding Deformation (50.0% conf.)<br>Overall Correlation (CCF): 0.6406   Max Dev: 34.17 dB |
|--------------------------------------------------------------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Sub-band Diagnostic Breakdown (IEEE C57.149 Standards Table)

| Sub-Band          | Frequency Range | Normalized Magnitude | Phase Shift (dB) | Health Status     |
|-------------------|-----------------|----------------------|------------------|-------------------|
| Low (Core)        | < 2 kHz         | 0.8844               | 27.07 dB         | Danger            |
| Mid (Winding)     | 2 - 100 kHz     | -0.4420              | 34.17 dB         | Critical          |
| High (Insulation) | > 100 kHz       | N/A                  | N/A              | Insufficient Data |

## Frequency Response Bode Plot Comparison



## Actionable Recommendations & Maintenance Guidance

**Primary Assessment:** Significant frequency response shift! Immediate internal inspection of windings recommended.

**Mandatory Protocol:** Significant frequency response deviations detected. Immediately de-energize asset if in service. Perform winding resistance, excitation current, and dissolved gas analysis (DGA). Schedule internal tank inspection to confirm mechanical winding/core displacement.

### Component-Level Guidance (IEEE C57.149):

- Core (Low Band, < 2 kHz):** Severe deviation detected. Magnetic core deformation or core ground fault suspected. Check core ground current and excitation current.
- Winding (Mid Band, 2–100 kHz):** Severe deviation detected. Radial or axial mechanical winding deformation indicated. Perform DC winding resistance and leakage reactance tests.
- Insulation (High Band, > 100 kHz):** Insufficient data points (< 10 points) in this sub-band. Re-sweep frequency response.

**Deterministic Physics Floor:** As mandated by IEEE C57.149-2012 and IEC 60076-18, the diagnostic status is governed strictly by the worst-performing sub-band (Low: Core, Mid: Winding, High: Insulation). Machine Learning predictions provide fault signature classification and escalation without overriding physical limits.